Gmail

YASH AGRAWAL <micdt_2025010@iiitm.ac.in>

Regarding 3D Object Detection (M. Tech Project)

YASH AGRAWAL <micdt_2025010@iiitm.ac.in> Thu, Sep 10, 2026 at 4:55 PM
To: Gaurav Sharma <gsharma@iiitm.ac.in>

Please Find Attached Values, Figures Dictionary
1.

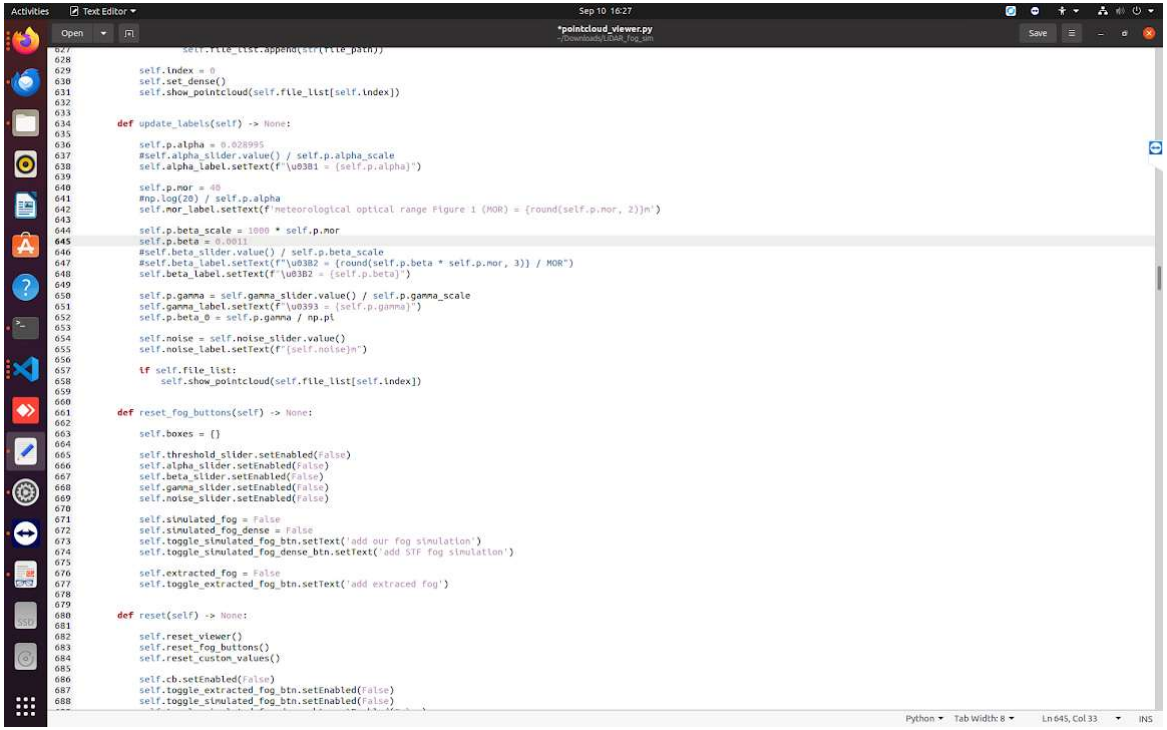
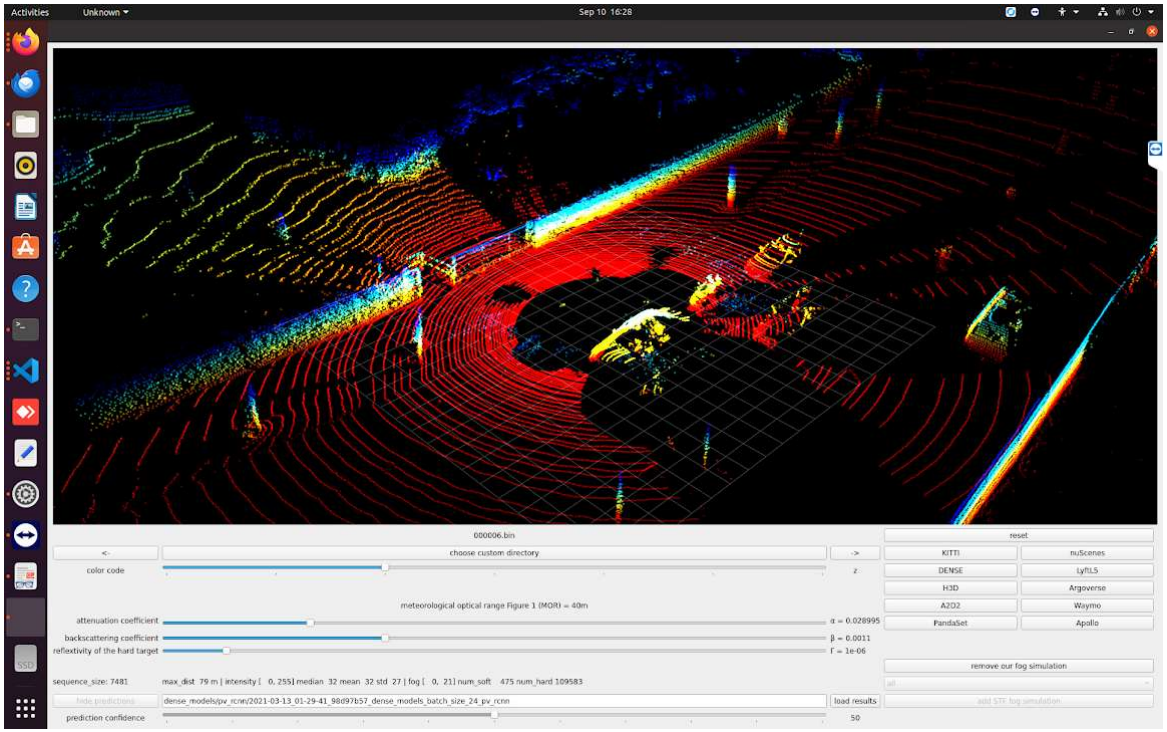


Figure (1):


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000006.bin
reset
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KITTI
nuScenes
color code
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DENSE
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Argoverse
meteorological optical range Figure 1 (MOR) = 40m
A2D2
Waymo
attenuation coefficient
α = 0.028995
PandaSet
Apollo
backscattering coefficient
β = 0.0011
reflextivity of the hard target
Γ = 1e-06
remove our fog simulation
sequence_size: 7481
max_dist 79 m | intensity [0, 255] median 32 mean 32 std 27 | fog [0, 21] num_soft 475 num_hard 109583
hide predictions
dense_models/pv_rcnn/2021-03-13_01-29-41_98d97b57_dense_models_batch_size_24_pv_rcnn
load results
prediction confidence
50

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2)

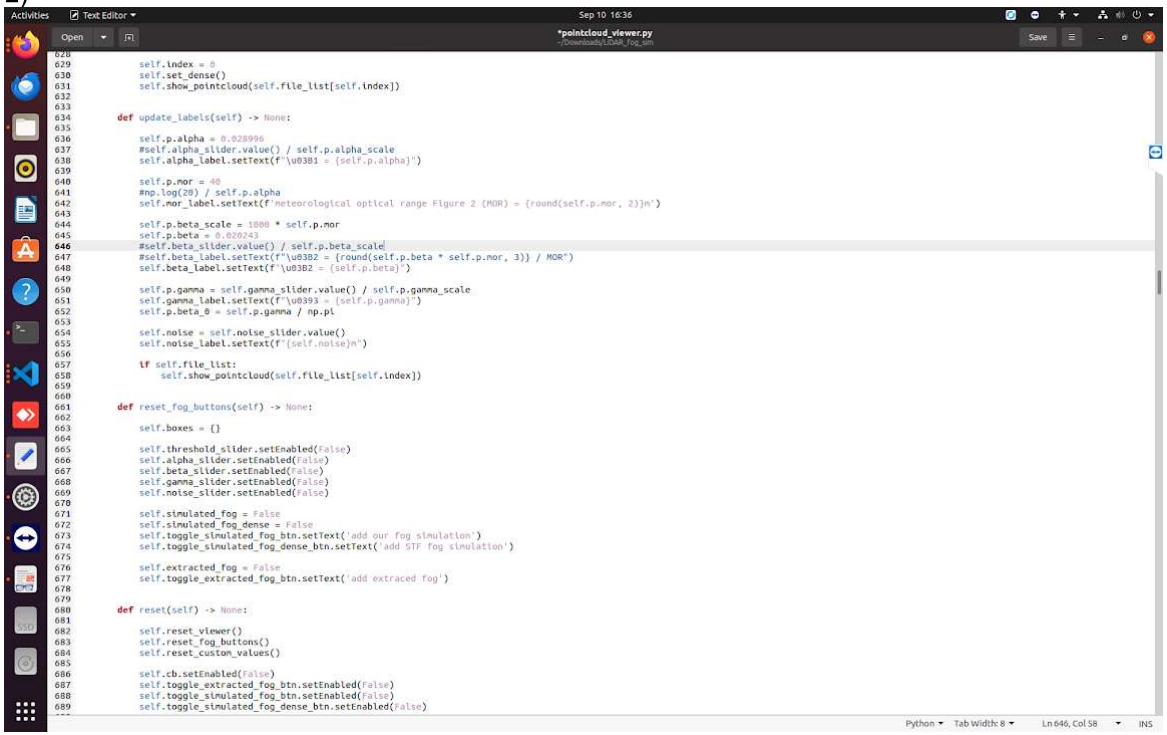
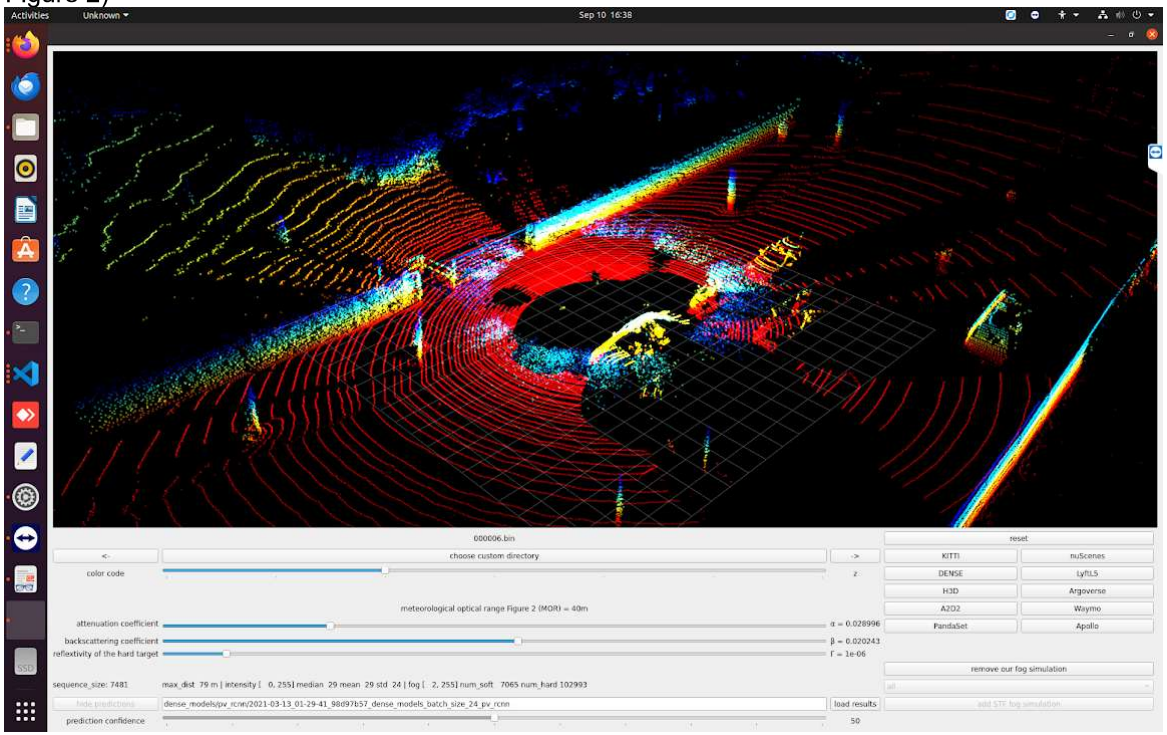

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self.index = 0
self.set_dense()
self.show_pointcloud(self.file_list[self.index])
def update_labels(self) -> None:
self.p.alpha = 0.028996
#self.alpha_slider.value() / self.p.alpha_scale
self.alpha_label.setText(f"\u03B1 = {self.p.alpha}")
self.p.mor = 40
#np.log(20) / self.p.alpha
self.mor_label.setText(f'meteorological optical range Figure 2 (MOR) = {round(self.p.mor, 2)}m')
self.p.beta_scale = 1000 * self.p.mor
self.p.beta = 0.020243
#self.beta_slider.value() / self.p.beta_scale
#self.beta_label.setText(f"\u03B2 = {round(self.p.beta * self.p.mor, 3)} / MOR")
self.beta_label.setText(f"\u03B2 = {self.p.beta}")
self.p.gamma = self.gamma_slider.value() / self.p.gamma_scale
self.gamma_label.setText(f"\u0393 = {self.p.gamma}")
self.p.beta_0 = self.p.gamma / np.pi
self.noise = self.noise_slider.value()
self.noise_label.setText(f"{self.noise}m")
if self.file_list:
self.show_pointcloud(self.file_list[self.index])
def reset_fog_buttons(self) -> None:
self.boxes = {}
self.threshold_slider.setEnabled(False)
self.alpha_slider.setEnabled(False)
self.beta_slider.setEnabled(False)
self.gamma_slider.setEnabled(False)
self.noise_slider.setEnabled(False)
self.simulated_fog = False
self.simulated_fog_dense = False
self.toggle_simulated_fog_btn.setText('add our fog simulation')
self.toggle_simulated_fog_dense_btn.setText('add STF fog simulation')
self.extracted_fog = False
self.toggle_extracted_fog_btn.setText('add extraced fog')
def reset(self) -> None:
self.reset_viewer()
self.reset_fog_buttons()
self.reset_custom_values()
self.cb.setEnabled(False)
self.toggle_extracted_fog_btn.setEnabled(False)
self.toggle_simulated_fog_btn.setEnabled(False)
self.toggle_simulated_fog_dense_btn.setEnabled(False)
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Figure 2)


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A2D2
Waymo
PandaSet
Apollo
meteorological optical range Figure 2 (MOR) = 40m
attenuation coefficient
backscattering coefficient
reflextivity of the hard target
remove our fog simulation
load results
prediction confidence

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3)

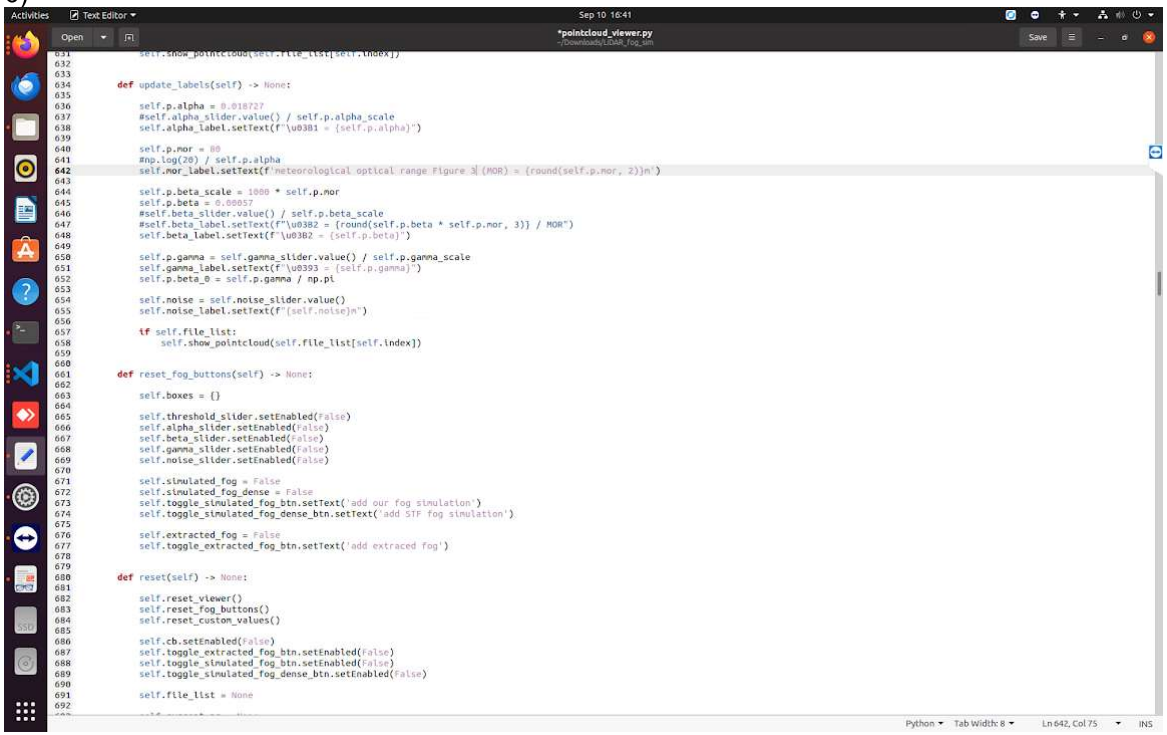
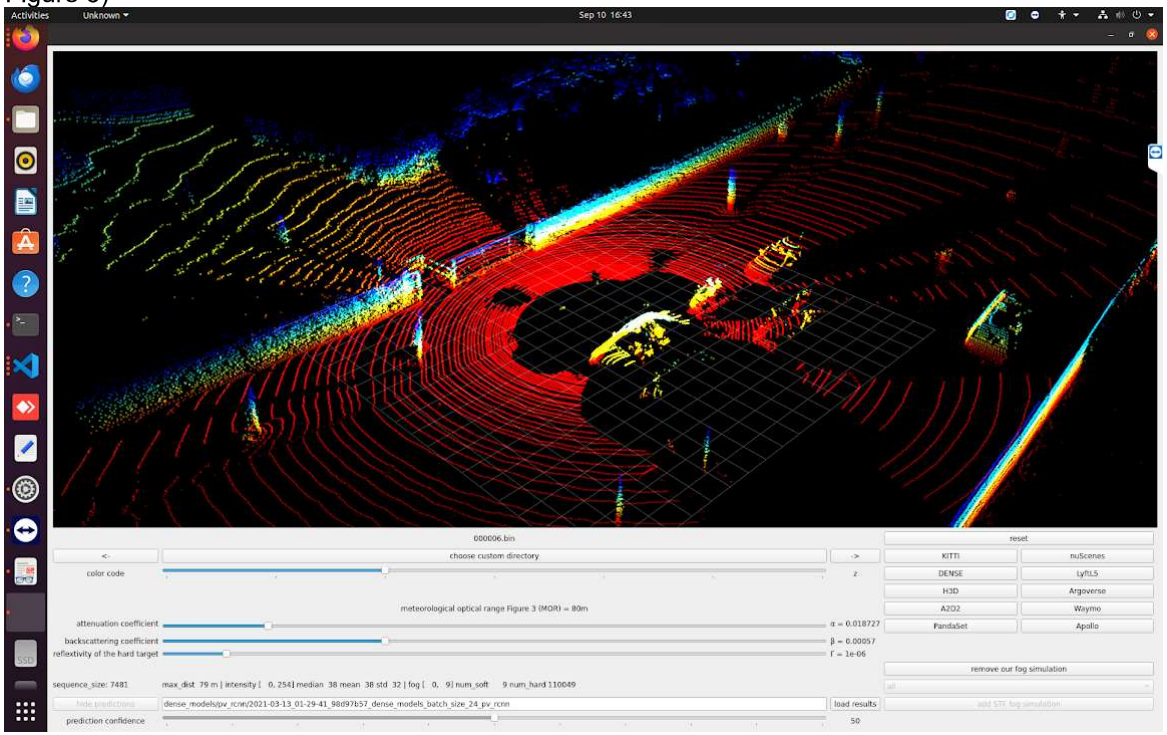

Activities
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self.show_pointcloud(self.file_list[self.index])
def update_labels(self) -> None:
self.p.alpha = 0.018727
#self.alpha_slider.value() / self.p.alpha_scale
self.alpha_label.setText(f"\u03B1 = {self.p.alpha}")
self.p.mor = 80
#np.log(20) / self.p.alpha
self.mor_label.setText(f'meteorological optical range Figure 3 (MOR) = {round(self.p.mor, 2)}m')
self.p.beta_scale = 1000 * self.p.mor
self.p.beta = 0.00057
#self.beta_slider.value() / self.p.beta_scale
#self.beta_label.setText(f"\u03B2 = {round(self.p.beta * self.p.mor, 3)} / MOR")
self.beta_label.setText(f"\u03B2 = {self.p.beta}")
self.p.gamma = self.gamma_slider.value() / self.p.gamma_scale
self.gamma_label.setText(f"\u0393 = {self.p.gamma}")
self.p.beta_0 = self.p.gamma / np.pi
self.noise = self.noise_slider.value()
self.noise_label.setText(f"{self.noise}m")
if self.file_list:
self.show_pointcloud(self.file_list[self.index])
def reset_fog_buttons(self) -> None:
self.boxes = {}
self.threshold_slider.setEnabled(False)
self.alpha_slider.setEnabled(False)
self.beta_slider.setEnabled(False)
self.gamma_slider.setEnabled(False)
self.noise_slider.setEnabled(False)
self.simulated_fog = False
self.simulated_fog_dense = False
self.toggle_simulated_fog_btn.setText('add our fog simulation')
self.toggle_simulated_fog_dense_btn.setText('add STF fog simulation')
self.extracted_fog = False
self.toggle_extracted_fog_btn.setText('add extraced fog')
def reset(self) -> None:
self.reset_viewer()
self.reset_fog_buttons()
self.reset_custom_values()
self.cb.setEnabled(False)
self.toggle_extracted_fog_btn.setEnabled(False)
self.toggle_simulated_fog_btn.setEnabled(False)
self.toggle_simulated_fog_dense_btn.setEnabled(False)
self.file_list = None
Python
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Figure 3)


Activities
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Argoverse
meteorological optical range Figure 3 (MOR) = 80m
A2D2
Waymo
attenuation coefficient
α = 0.018727
PandaSet
Apollo
backscattering coefficient
β = 0.00057
reflextivity of the hard target
Γ = 1e-06
remove our fog simulation
sequence_size: 7481
max_dist 79 m | intensity [0, 254] median 38 mean 38 std 32 | fog [0, 9] num_soft 9 num_hard 110049
hide predictions
dense_models/pv_rcnn/2021-03-13_01-29-41_98d97b57_dense_models_batch_size_24_pv_rcnn
load results
prediction confidence
50

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4)

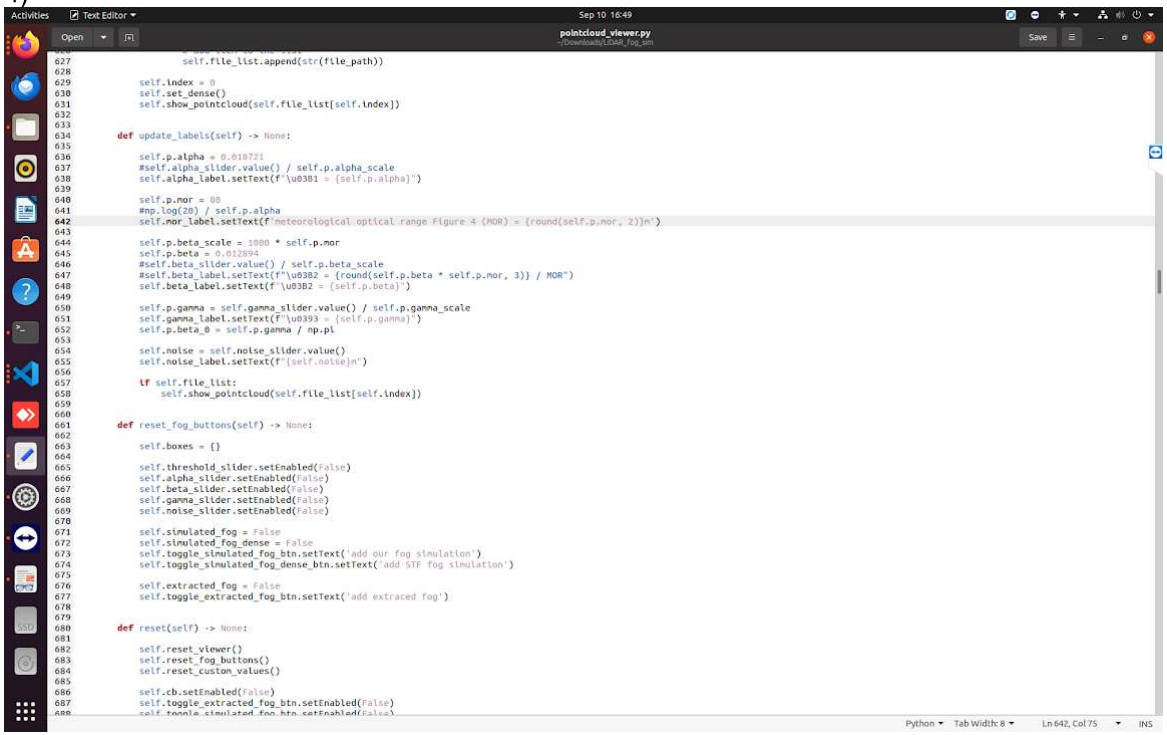
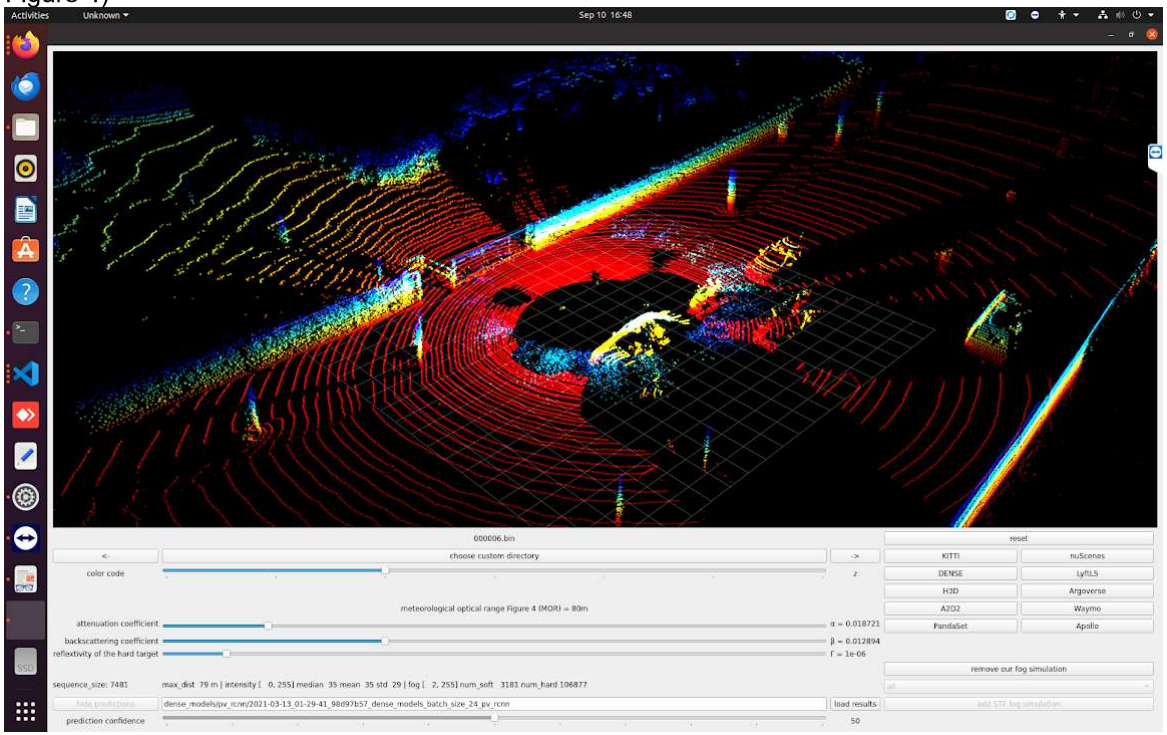

Activities
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Sep 10 16:49
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pointcloud_viewer.py
~/Downloads/LiDAR_Fog_Sim
Save
627 self.file_list.append(str(file_path))
628
629 self.index = 0
630 self.set_dense()
631 self.show_pointcloud(self.file_list[self.index])
632
633
634 def update_labels(self) -> None:
635
636 self.p.alpha = 0.018721
637 #self.alpha_slider.value() / self.p.alpha_scale
638 self.alpha_label.setText(f"\u03B1 = {self.p.alpha}")
639
640 self.p.mor = 80
641 #np.log(20) / self.p.alpha
642 self.mor_label.setText(f'meteorological optical range Figure 4 (MOR) = {round(self.p.mor, 2)}m')
643
644 self.p.beta_scale = 1000 * self.p.mor
645 self.p.beta = 0.012894
646 #self.beta_slider.value() / self.p.beta_scale
647 #self.beta_label.setText(f"\u03B2 = {round(self.p.beta * self.p.mor, 3)} / MOR")
648 self.beta_label.setText(f"\u03B2 = {self.p.beta}")
649
650 self.p.gamma = self.gamma_slider.value() / self.p.gamma_scale
651 self.gamma_label.setText(f"\u0393 = {self.p.gamma}")
652 self.p.beta_0 = self.p.gamma / np.pi
653
654 self.noise = self.noise_slider.value()
655 self.noise_label.setText(f"{self.noise}m")
656
657 if self.file_list:
658 self.show_pointcloud(self.file_list[self.index])
659
660
661 def reset_fog_buttons(self) -> None:
662
663 self.boxes = {}
664
665 self.threshold_slider.setEnabled(False)
666 self.alpha_slider.setEnabled(False)
667 self.beta_slider.setEnabled(False)
668 self.gamma_slider.setEnabled(False)
669 self.noise_slider.setEnabled(False)
670
671 self.simulated_fog = False
672 self.simulated_fog_dense = False
673 self.toggle_simulated_fog_btn.setText('add our fog simulation')
674 self.toggle_simulated_fog_dense_btn.setText('add STF fog simulation')
675
676 self.extracted_fog = False
677 self.toggle_extracted_fog_btn.setText('add extraced fog')
678
679
680 def reset(self) -> None:
681
682 self.reset_viewer()
683 self.reset_fog_buttons()
684 self.reset_custom_values()
685
686 self.cb.setEnabled(False)
687 self.toggle_extracted_fog_btn.setEnabled(False)
Python
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Figure 4)


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meteorological optical range Figure 4 (MOR) = 80m
A2D2
Waymo
attenuation coefficient
α = 0.018721
PandaSet
Apollo
backscattering coefficient
β = 0.012894
reflextivity of the hard target
Γ = 1e-06
remove our fog simulation
sequence_size: 7481
max_dist 79 m | intensity [0, 255] median 35 mean 35 std 29 | fog [2, 255] num_soft 3181 num_hard 106877
all
hide predictions
dense_models/pv_rcnn/2021-03-13_01-29-41_98d97b57_dense_models_batch_size_24_pv_rcnn
load results
add STF fog simulation
prediction confidence
50

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5)

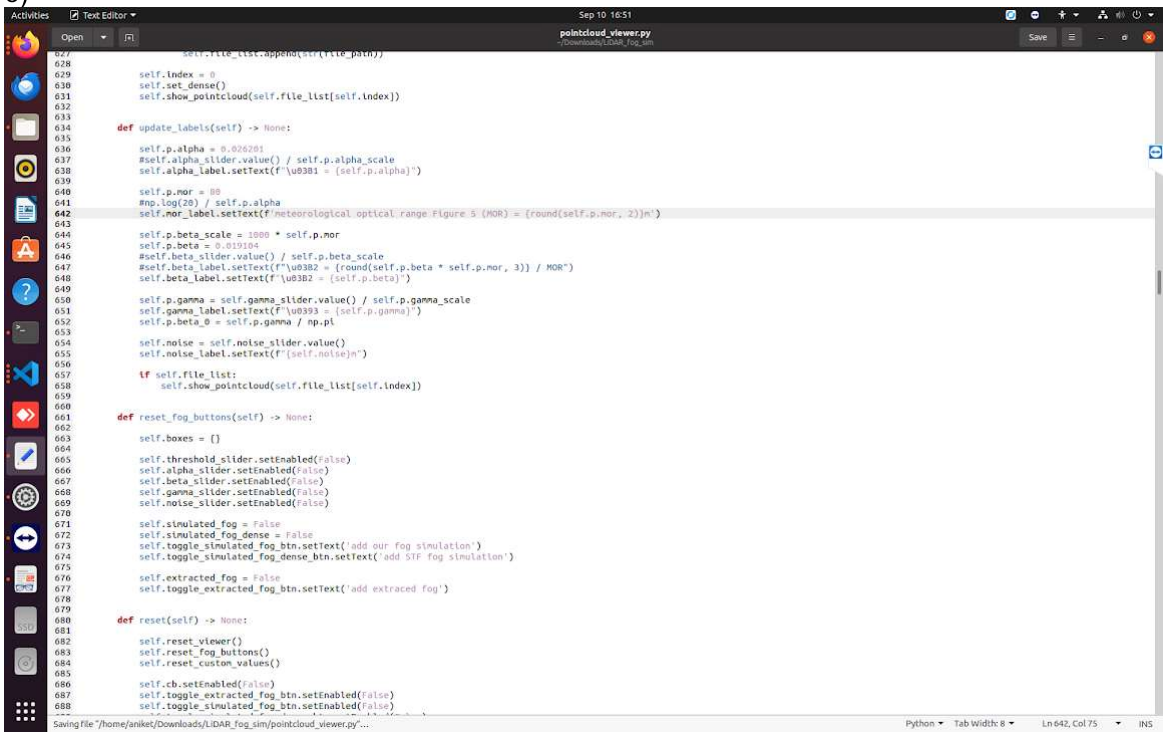
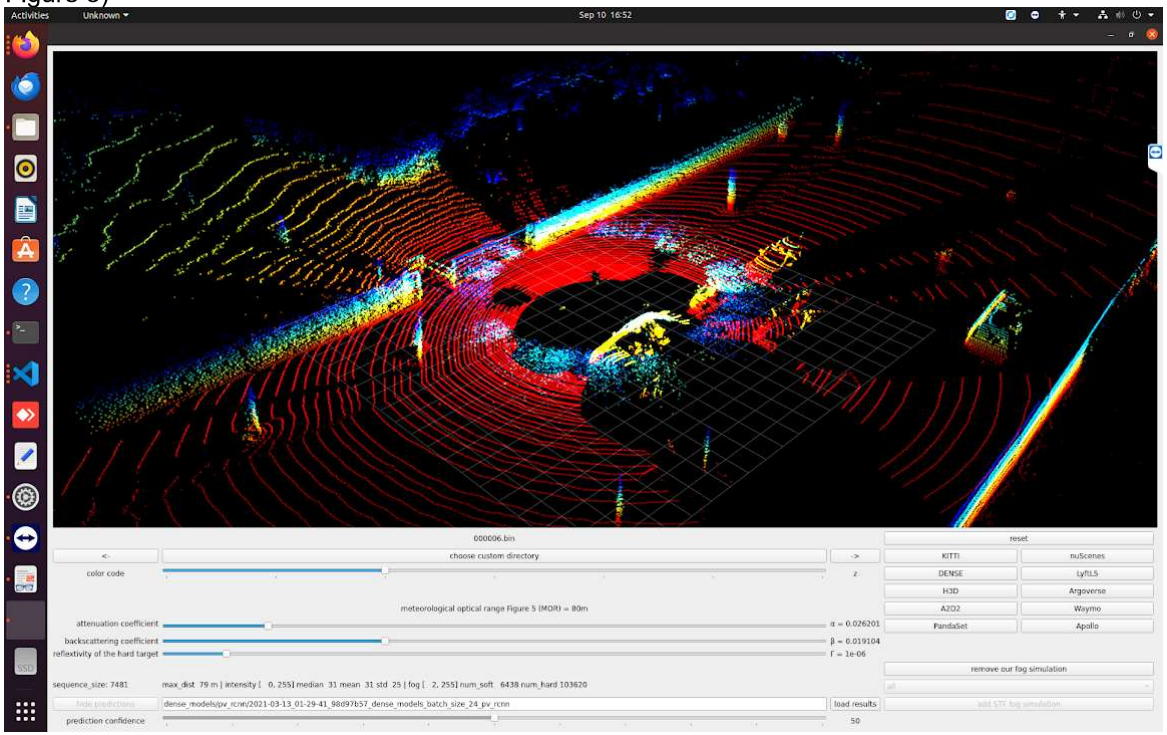
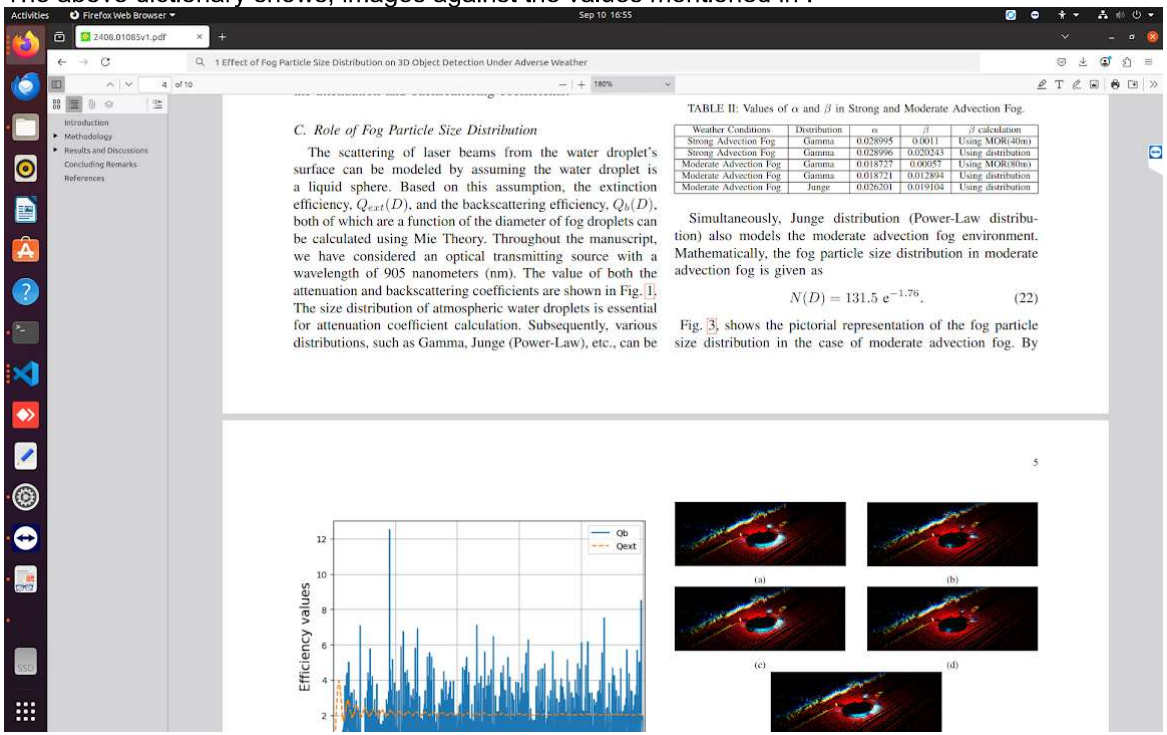

Activities
Text Editor
Sep 10 16:51
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self.file_list.append(str(file_path))
self.index = 0
self.set_dense()
self.show_pointcloud(self.file_list[self.index])
def update_labels(self) -> None:
self.p.alpha = 0.026201
#self.alpha_slider.value() / self.p.alpha_scale
self.alpha_label.setText(f"\u03B1 = {self.p.alpha}")
self.p.mor = 80
#np.log(20) / self.p.alpha
self.mor_label.setText(f'meteorological optical range Figure 5 (MOR) = {round(self.p.mor, 2)}m')
self.p.beta_scale = 1000 * self.p.mor
self.p.beta = 0.019104
#self.beta_slider.value() / self.p.beta_scale
#self.beta_label.setText(f"\u03B2 = {round(self.p.beta * self.p.mor, 3)} / MOR")
self.beta_label.setText(f"\u03B2 = {self.p.beta}")
self.p.gamma = self.gamma_slider.value() / self.p.gamma_scale
self.gamma_label.setText(f"\u0393 = {self.p.gamma}")
self.p.beta_0 = self.p.ganma / np.pi
self.noise = self.noise_slider.value()
self.noise_label.setText(f"{self.noise}n")
if self.file_list:
self.show_pointcloud(self.file_list[self.index])
def reset_fog_buttons(self) -> None:
self.boxes = []
self.threshold_slider.setEnabled(False)
self.alpha_slider.setEnabled(False)
self.beta_slider.setEnabled(False)
self.gamma_slider.setEnabled(False)
self.noise_slider.setEnabled(False)
self.simulated_fog = False
self.simulated_fog_dense = False
self.toggle_simulated_fog_btn.setText('add our fog simulation')
self.toggle_simulated_fog_dense_btn.setText('add STF fog simulation')
self.extracted_fog = False
self.toggle_extracted_fog_btn.setText('add extraced fog')
def reset(self) -> None:
self.reset_viewer()
self.reset_fog_buttons()
self.reset_custom_values()
self.cb.setEnabled(False)
self.toggle_extracted_fog_btn.setEnabled(False)
self.toggle_simulated_fog_btn.setEnabled(False)
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Python
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Figure 5)


Activities
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color code
DENSE
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meteorological optical range Figure 5 (MOR) = 80m
A2D2
Waymo
attenuation coefficient
PandaSet
Apollo
backscattering coefficient
reflextivity of the hard target
remove our fog simulation
sequence_size: 7481
load results
prediction confidence

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The above dictionary shows, images against the values mentioned in :



1 Effect of Fog Particle Size Distribution on 3D Object Detection Under Adverse Weather

Introduction
Methodology
Results and Discussions
Concluding Remarks
References

C. Role of Fog Particle Size Distribution

The scattering of laser beams from the water droplet's surface can be modeled by assuming the water droplet is a liquid sphere. Based on this assumption, the extinction efficiency, $Q_{ext}(D)$, and the backscattering efficiency, $Q_b(D)$, both of which are a function of the diameter of fog droplets can be calculated using Mie Theory. Throughout the manuscript, we have considered an optical transmitting source with a wavelength of 905 nanometers (nm). The value of both the attenuation and backscattering coefficients are shown in Fig. 1. The size distribution of atmospheric water droplets is essential for attenuation coefficient calculation. Subsequently, various distributions, such as Gamma, Junge (Power-Law), etc., can be

TABLE II: Values of α and β in Strong and Moderate Advection Fog.

Weather Conditions	Distribution	α	β	β calculation
Strong Advection Fog	Gamma	0.028995	0.0011	Using MOR(40m)
Strong Advection Fog	Gamma	0.028996	0.020243	Using distribution
Moderate Advection Fog	Gamma	0.018727	0.00057	Using MOR(80m)
Moderate Advection Fog	Gamma	0.018721	0.012894	Using distribution
Moderate Advection Fog	Junge	0.026201	0.019104	Using distribution

Simultaneously, Junge distribution (Power-Law distribution) also models the moderate advection fog environment. Mathematically, the fog particle size distribution in moderate advection fog is given as

$$N(D) = 131.5\ e^{-1.76}. \quad (22)$$

Fig. 3, shows the pictorial representation of the fog particle size distribution in the case of moderate advection fog. By

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Thank YOu!

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